

 <https://github.com/ColitoBandito/cs4412-project-Collin>

Mapping Mood Profiles and Feature Associations of Spotify Tracks

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Data Mining W01

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1 INTRODUCTION

1.1 Project Overview and Motivation

In the landscape of modern music, traditional genre labels are becoming increasingly less sufficient in describing the nuances of a particular track. While genres such as "Pop" or "Rock" give people a broad category to associate music with, they fail to capture the particular "audio fingerprint" that is more so responsible for giving the listener an emotional response. The motivation behind this project is to use a variety of Data Mining techniques in order to determine truths past these static labels, unveiling "Mood Profiles" or groups of tracks that share mathematically consistent audio attributes. Hopefully, through finding these patterns within the objective data of over 130,663 tracks sourced from Spotify, some groundwork can be laid for improvement on music recommendation engines that provide track suggestions less based on genre labels/genre preferences and more so on a user's emotional state.

1.2 Discovery Questions

To help guide both the technical groundwork and eventual result evaluations, this project included three primary discovery questions:

- Audio Fingerprinting: What distinct clusters of tracks exist based purely on objective audio fingerprints?
- Feature Drivers: Which specific audio attributes, such as the inverse relationship between Acousticness and Energy, act as the primary drivers of song similarity?
- Structural Density: What does the 2D projection reveal about the density and separation of musical mood structures?

1.3 Key EDA Insights

This initial phase of exploratory analysis has shown a number of patterns that give an idea how to approach the mining phase of this project.

- Strong Feature Correlations: A high positive correlation (0.77) was found between Loudness and Energy, while there was a significant negative correlation (-0.71 between acousticness and energy).

- **Data Density:** Early scatter plots showed that Valence and energy had a dense, overlapping distribution across all features. This confirmed that algorithmic clustering is helpful in distinguishing "mood profiles", beyond just regular listener intuition.
- **Feature Skew:** Attributes such as Instrumentalness skewed strongly towards zero, which indicates that, at least within this dataset, the majority of tracks were vocal-centric tracks.

2 METHODS AND PREPROCESSING

2.1 Data Integrity and Cleaning

Data integrity was paramount in ensuring proper behavior of distance-based mining algorithms. I employed an early check for null values and that check revealed zero missing entries, ensuring the soon to come distance calculations would not be hurt by incomplete vectors. All duplicate entries were then erased as the same track could be featured on multiple albums. This was to prevent duplicates that could skew data density and give false impressions on the importance of certain audio profiles.

2.2 Standardization via StandardScaler

Some attributes such as Tempo (BPM) and Loudness (db) existed on much larger scales than others, with Valence being only on a scale of (0.0-1.0). Without employing standardization, some algorithms such as K-Means would decide to over prioritize the exponentially larger attributes. As such, all numerical attributes were standardized using StandardScaler to eliminate any potential magnitude bias. This gave assurance that a track's "happiness" score, or Valence, would have an equal vote in clustering as its tempo.

2.3 Dimensionality Reduction

Principal Component Analysis (PCA) was used to project that initial 11-dimensional space into two useful pieces, ultimately capturing 55.1 percent of total variance. This did enable 2D visualizations, but it also enabled identifying key data drivers. PC1 was the Intensity Axis, capturing 40.3 variance, and it's driven by the inverse relationship between Acousticness and Energy/Loudness. It served to separate Organic music from electronically produced music. PC2, the Positivity Axis, captured 14.8 variance and is driven by Valence and Danceability. It mostly tracked the movement between Dark/Sad sounds to Happy.

3 MINING RESULTS

The mining phase of this project was primarily centered on finding repeatable patterns within the high dimensionality space of my track dataset. It included three distinct phases of discovery: cluster optimization, centroid mapping and hierarchical rule auditing.

3.1 Cluster Optimization and Mathematical Validation

Determining the optimal number of clusters (K) required for the model to effectively answer my discovery questions took a balance of simplicity and depth. The number that was ultimately decided upon, four, was found through a dual-validation approach using both the Elbow Method and Silhouette Analysis.

- Inertia and the Elbow Method: The Within-Cluster Sum of Squares (WCSS) was calculated for k values ranging from 1 to 10. A distinct inflection point appeared at k=4, where the rate of decrease in inertia began to level off, indicating a point of diminishing returns for added complexity.
- To mathematically defend this decision, Silhouette scores were calculated for k=2 through k=5. While k=2 yielded the highest mean silhouette coefficient of 0.336, indicating the most distinct density split, k=4 maintained a defensible score of 0.181. This provided a higher level of granularity required for the discovery questions while ensuring that clusters remained more distinct than the k=5 configuration (0.176).

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3.2 Quantitative Cluster Centroids and Mood Profiles

The K-Means algorithm identified four stable centroids within the 11-dimensional feature space. The following table represents the mean values for each audio attribute across the standardized dataset, providing a mathematical "fingerprint" for each discovered profile.

Table 1: Discovered Spotify Mood Profiles ($k = 4$)

Cluster	Key Characteristics (Centroids)	Proposed Mood Label
0	High Valence (0.68), High Danceability (0.70)	Party
1	High Acousticness (0.84), Low Energy (0.20)	Acoustic
2	High Energy (0.67), High Loudness (-7.51 dB)	Dark
3	Highest Speechiness (0.36), High Danceability (0.68)	Lyrical

Cluster 0 (Party): Representative of positive-sounding music. High valence suggests the music here contains an optimistic tone that is easy to dance to and reminiscent of modern Pop. **Cluster 1 (Acoustic):** The most distinct of all profiles generated; it shows a strong inverse relationship between acousticness and energy. This cluster represents "unplugged" tracks with a lower tempo. **Cluster 2 (Dark):** Energetic but not "happy" music sonically. This grouping contains music with more serious tones and includes tracks where high intensity is not associated with a positive mood. **Cluster 3 (Lyrical):** Contains tracks that are on average five times more "wordy" than other groups. This encompasses rap and hip-hop where the vocal delivery is performed prominently over the instrumentation.

3.3 Decision Tree Audit

To get past the blurriness that occurred on the previous PCA visualization, I implemented a decision tree that acts as a sort of discovery tool. As opposed to PCA, which reduces data, this tree views all features with their original dimensions to create exact, logical branches between music segments. This gives a hierarchy of musical attributes that distinguish mood profiles. The first key discovery from the tree is, surprisingly, that Valence is the primary root split, followed by energy and Speechiness. Initially I thought the primary branch would be acousticness, but valence happens to be more of a bridge to the acoustic cluster as low valence (< 0.33) almost exclusively leads to it. There's a Sad/Deep split that occurs if a song is < 0.47 valence where it then looks at Energy. Higher energy leads to the Dark cluster, indicating that Dark music is moreso defined by intensity than emotional positivity. Also noteworthy is the "Vibe" split where if a song is Happy (Valence > 0.47), the model looks at Speechiness. Low Speechiness then defines the Party cluster where valence is (< 0.24), showing a focus on danceability and rhythm. High Speechiness is, as expected, the absolute identifying trait of the Lyrical cluster.

4 INTERPRETATION AND DOMAIN SIGNIFICANCE

4.1 Musical Identity of the Clusters

The four identified mood profiles found with this model can act as distinct and usable states for a streaming platform's recommendation logic:

- Acoustic Profile (Cluster 1): Represents the organic extreme of music. Its high separation shows that acoustic instrumentation is the most reliable indicator of low-energy/low-loudness in audio. In a business or casual use context, this could lend to recommendations of "Deep Work" or "Study" contexts.
- Dark Profile (Cluster 2): The significance of this discovery is in decoupling "Energy" from perceived happiness. These tracks are intense in terms of their sonics, but emotionally quite somber. This segment could cover tracks that are used for high intensity focused activities or workouts where there isn't necessarily a melodic resolution in the track. This is seen in sub-genres like deep electronic or heavy metal.
- The "Party" vs "Lyrical" Split: Clusters 0 and 3 are tied together by high Valence, or musical positivity, but are split by the volume of vocals on the tracks. The Lyrical cluster shows music that is narrative heavy, such as hip hop, where lyrical more so identifies catchy and rhythmic tracks.

4.2 Positivity First

What is likely the most profound insight is derived from the Decision Tree audit where Valence was discovered to be the root decision node. This indicates that human perception of music is taken first and foremost through a lens of positivity. This perception takes precedence before instrumentation or volume levels. For a recommendation engine, this is useful in that matching a user's current "vibe" is more important than matching a stated preferred genre or type of instrumentation.

5 CRITICAL ASSESSMENT

5.1 Validity and Robustness

In order to be sure that findings were not delusions of algorithmic assumptions, cross validation occurred using two different methods: Stability testing with multiple K-means seeds and a DBSCAN density check. The K-Means algorithm was executed across five different and random seeds, whereby resulting centroids remained consistent and with statistically irrelevant shifts in mean values. This gives credence that the decision for four clusters is stable. The density-based check found a highly dense core of 83.3 of the tracks. This mostly just confirmed that that Spotify dataset is a continuous cloud. While the mood profiles generated are within rigid boundaries, they are in actuality granular pieces of these high density regions.

5.2 Technical and Scope Limitations

I acknowledge two primary limitations of my analysis: an assumption of cluster geometry with K-Means and a lack of lyrical context. Firstly, K-Means by nature assumes clusters are spherical in shape and roughly equal in size. The actuality of my data in this project was a continuous and asymmetrical cloud. Tracks that land on the borders of this cloud represent more fluid emotional states that a hard-partitioning algorithm like K-Means may opt to oversimplify. Secondly, this study is focused on "audio fingerprints" and therefore cannot account for any irony between the sonic attributes of a song and the lyrics within it. An example would be Pumped Up Kicks, where it's conventionally listened to as a party track but the lyrics are anything but cheerful.

5.3 Ethical Considerations

Applying emotional profiling to a pool of 130,000+ tracks assumes two primary ethical risks: emotional echo chambers and a bias towards Western metrics. Firstly, Mood based recommendation could accidentally cause a negative feedback loop. If an algorithm determines low valence music ought to be repeatedly shared with a user in a poor emotional state, it may continually reinforce those negative emotions to an unhealthy degree. Secondly, the audio attributes of Valence and Danceability are most often tuned to Western Musical Theory standards. These Western influenced metrics may fail in profiling joyous signatures from other non-western cultures. In doing so, this could lead to algorithmic bias in how global music platforms categorize music.

6 EXECUTIVE SUMMARY

The aim of this project is to look into the objective "audio fingerprints" of over 130,000 Spotify tracks in order to categorize them differently than traditional genre labels. By using the data mining techniques of K-Means clustering and PCA, the study finds four distinct mood profiles: Party, Acoustic, Dark and Lyrical. An important finding, confirmed from a Decision Tree Audit, showed that Valence (musical positivity) is the primary root split for musical classification. This supposes how a song makes you respond emotional is the driver of that songs identity. While DBSCAN validation gives confirmation that all tracks belong in a dense, continuous cloud rather than rigid groupings, the identified mood profiles still give a solid framework for creating recommendation engines that are in alignment with a user's real time emotional state.

7 CONCLUSION

7.1 Summary of Findings

This study successfully mapped the underlying structure of Spotify's audio structure by moving past genre labels and into more data-driven classification. Through the use of unsupervised learning on audio features, distinct Mood Profiles emerged that reflect consistent patterns in how tracks are constructed. The analysis shows that Energy and Acousticness primarily define the physical intensity of a track, while Valence operates as the key driver of emotional interpretation. Together, these features form a reliable foundation for distinguishing how music both sounds and feels

7.2 Domain Implications

Being able to mathematically define these mood-based segments has potentially significant implications for streaming platforms. A shift from the traditional genre approach to a more mood centric framework allows recommendation systems to better align with user context. Instead of only matching historical preferences, platforms can begin to account for a user's current state or activity, whether that be studying, focusing, or socializing. This creates the potential for recommendation engines that are more adaptive or reflective of real time user intent.